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Diagnosis, Prediction, and Risk Factor Analysis of Multiple Sclerosis Disease using Quantum Machine Learning Classifier

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***Abstract*—Multiple sclerosis (MS) is a condition that affects the brain or central nervous system, which can result in issues with the optic nerve, brain, and spinal cord. Multiple sclerosis is thought to affect 2.8 million people. This research focuses on the diagnosis of multiple sclerosis using Quantum machine learning (QML) classifiers. Several models have been created utilizing various data sets, such as clinical and magnetic resonance imaging (MRI) data. The quantum machine learning has emerged by the merging of classical machine learning and quantum information processing as a result of the superiority and advancement of quantum computing in various fields (such as encryption, machine learning, and healthcare). The experimental results present a comparative analysis of classical machine learning and quantum machine learning algorithms for the problem under consideration.**

Keywords—Multiple Sclerosis, Variational Quantum Classifier, Quantum Machine Learning, Quantum Support Vector Classifier, Diagnosis, Prediction, Risk Factor Analysis, Qiskit.

I. Introduction

An autoimmune-mediated condition that impacts the central nerve system (CNS) is called multiple sclerosis (MS) which is the most common neurological disorder. The youth today, is most frequently affected by neurological issues such as MS as well as severe physical or cognitive incapacitation.[1].The multifocal inflammatory zones caused by infiltrations of macrophages and focal T-lymphocytic and oligodendrocyte fatal conditions are the primary reasons why myelin sheaths break down. This causes CNS plaques to form which are made up of cells that cause inflammation and their products, the demyelinated and transacted axons in both the grey and white matter astrocytes [2]. The proper transmission of nerve impulses may be interfered with by these lesions, which can result in neuronal dysfunction including autonomic and sensorimotor disorders, visual disturbances, ataxia, exhaustion, cognitive challenges, and emotional issues [3]. The subtypes of multiple sclerosis known as relapsing-remitting MS (RRMS), progressive relapsing multiple sclerosis (PRMS), primary progressive multiple sclerosis (PPMS), secondary progressive multiple sclerosis (SPMS), and are considered to be important in terms of prognosis and available treatments [4]. The most prevalent subtype, accounting for roughly 87% of cases, is RRMS, which is typified by unpredictable acute episodes interspersed by remissions [5]. Inflammatory assaults on myelin and nerve fibers take place during RRMS. Lesions in the CNS brought on by activated immune cells result in symptoms such as learning and memory impairment, tingling and numbness, intermittent episodes of exhaustion, digestive and urinary system problems, spasticity, and visual impairments [6]. About 10–15% of MS patients are diagnosed with PPMS, which mostly affects the spinal cord's nerves.

A precise and correct diagnosis of MS is required in order to facilitate the early treatment of the condition. Since the medications that modify the disease help manage symptoms and halt the disease from progressing [7]. The basis of the MS diagnosis is the existence of CNS abnormalities that differ from each other in time and space, as well as the exclusion of any other diseases that may mimic multiple sclerosis both radiologically and clinically [8].

To date, The most efficient method for diagnosing MS, tracking the progression of the condition, and evaluating the efficacy of various treatment options in clinical trials is magnetic resonance imaging (MRI) [9]. The laborious and time-consuming MRI-based MS diagnosis procedure is prone to human error. As a result, quantum machine learning is being applied to automate MS diagnosis [10].

This study investigates the use of QML and demonstrates a noteworthy advantage in improving multiple sclerosis diagnosis and prediction. We aim to compare Quantum Support Vector Classifier (QSVC) and Variational Quantum Classifier (VQC), two Quantum classification techniques. Quantum Kernels and Quantum Machine Learning—two essential computational building blocks that we employ in the classification of multiple sclerosis—were built using Qiskit Machine Learning.

II. LITERATURE REVIEW

Numerous characteristics of quantum machines remain unclear despite various methods of investigation. Two basic approaches to QML research may be distinguished: kernel-based and quantum neural network (QNN)-based. The foundation for utilizing a quantum algorithm for a kernel-based approach was provided in [11]. Like a high dimensional kernel, they concentrated on the fundamental law of quantum mechanics in Hilbert space. They discovered mathematical connections between quantum computing, kernel techniques, and feature maps. Additionally, this work carried out a real experiment that builds basic benchmark classifiers employing a support vector machine kernel based on a quantum feature map. Practical research aimed at exploring and developing QSVM has surfaced recently.

Applying appropriate feature maps can enhance model efficiency by experimentation with different feature maps, as demonstrated in [12]. When QSVM was applied to sentimental classification tasks, it outperformed classical methods, as demonstrated in [13]. This suggested that kernel-based QML could find applications in natural language processing (NLP) applications. Hybrid-QSVM was utilized to recognize clouds in satellite images with many wavelengths. Using simple quantum kernels on huge datasets instead of intricate entangled circuits, they saw great accuracy. The alternative viewpoint entails applying traditional deep learning principles to variational quantum circuit (VQC) architecture. Numerous trainable parameterized rotation gates with the ability to compute gradients or backpropagate in a typical manner make up VQC [14].

In order to evaluate the conventional and quantum machine learning-based computer-aided diagnostics for the identification of skin cancer, a statistical performance analysis was conducted, as described in [15]. In this case, a traditional SVM method beat the quantum support vector machine (QSVM) approach. The experiment was conducted using a single feature extraction technique, according to the authors. Therefore, it's possible that using quantum-aided feature-fusion will increase the accuracy of skin cancer detection.

In [16] the authors consider the increased training accuracy as a sign that QNN will perform better in the future. To that end, they intend to use Monte Carlo simulations to statistically validate the first best-case results, adjust hyperparameters, and implement quantum noise error mitigation strategies. The authors presented a binary class classification model using Quantum Deep Networks.

III. PROPOSED METHODOLOGY

To develop an efficient model for categorizing data involved data collection and preprocessing. As mentioned in section II the dataset was gathered from Kaggle. The study mainly focused on Mexican patients' diagnoses with the CIS dataset. The dataset is mainly prepared for modeling and analysis.

Further subsections detail the methodology like data preprocessing. The study comprised dimensionality reduction and normalization. Subsequently, the subsection describes the use of ansatz for parameter optimization and the design of feature maps.

A. Data Collection

The dataset was collected from Kaggle. This data came from a study conducted between 2006 and 2010 on mestizo Mexican patients who visited the National Institute of Neurology and Neurosurgery in Mexico after being diagnosed with CIS. The performance results for the conventional machine learning method SVM, which was derived from the number of features extracted using the feature extraction technique, are shown in Table 1.

TABLE I. FEATURE DESCRIPTION OF COLLECTED DATASET

Feature	Description	Range
ID	Patient identifier	-
Age:	Patient’s age in years	0-80
Schooling	Time in years spent by the patient in school.	10-25
Gender	Male/Female	1/2
Breastfeeding	Yes/No/Unknown	1-3
Varicella	Varicella, also known as chicken pox or varicella zoster virus (VZV), is a highly infectious disease that is initially caused by an infection with a member of the herpesvirus family.	1=positive, 2=negative, 3=unknown
Initial_Symptoms:	Visual, motor, sensory, and other; tactile, motor, and other.	1-15
Mono _or_Polysymptomatic	Polysymptomatic/Monosymptomatic/ Unknown	1-3
Oligoclonal_Bands	Oligoclonal bands (OCBs) are immunoglobulin bands that are detected, upon investigation, in cerebrospinal fluid or blood serum from a patient.	0=negative, 1=positive, 2=unknown
LLSSEP	After stimulating peripheral nerves, the central nervous system records somatosensory evoked potentials, or SSEPs. ULSSEP (upper limb SSEP)	0=negative, 1=positive
ULSSEP	LLSSEP (lower limb SSEP)	0=negative, 1=positive
VEP	An evoked potential known as a visual evoked potential (VEP) can be utilized to verify injury to the occipital cortex, optic nerve, optic chiasm, optic radiations, and retina.	0=negative, 1=positive
BAEP	Brainstem auditory evoked responses (BAERs) are minute auditory evoked potentials triggered by sound stimuli that are captured by scalp electrodes.	0=negative, 1=positive
Periventricular_MRI	Negative/Positive	0/1
Cortical_MRI	Negative/Positive	0/1
Infratentorial_MRI	Negative/Positive	0/1
Spinal_Cord_MRI	Negative/Positive	0/1
initial_EDSS	A technique for measuring disability in multiple sclerosis and tracking	1-3

Feature	Description	Range
	changes in disability over time is the Expanded Disability Status Scale (EDSS).	
final_EDSS	Final The Expanded Disability Status Scale (EDSS)	1-3
Group	CDMS/Non-CDMS	1/2

B. Data Preprocessing

- Feature Normalization - The next step in the data preprocessing phase is to normalize the features. In this instance, all features are squeezed onto the interval [0,1]. The values derived from additional QML variations, such as Real Amplitude and EfficientSU2, are also shown in Table 1. To do this, we can utilize scikit-learn's MinMaxScaler. This accomplishes exactly what is needed without the need to give parameters: data is mapped onto [0,1].
- Feature Reduction - Prior to model building, the amount of features in our dataset is reduced. This is accomplished by using PCA transformation techniques. Our 14 features are reduced to just 9 features. This decrease in dimensionality is solely intended for researchers' experimentation.

C. Model Building

A variational quantum classifier (VQC) is used to train the quantum machine learning model. The VQC is the most basic classifier offered by Qiskit Machine Learning. The VQC stands for "variational quantum classifier." It builds an automated quantum neural network given an ansatz and a feature map. Generally, building a legitimate classifier only requires passing the quantity of qubits and a quantum instance.

A method for encoding the data as qubits is required. This procedure is essential to obtaining a functional quantum model. This mapping is commonly referred to as data encoding, data embedding, or data loading, and the feature map plays a part in it.

One of the common feature maps in the Qiskit circuit library is the ZZFeatureMap. The feature map will contain num_features or 4 qubits since we pass num_features as the feature dimension. Fig. 1 Represents the quantum circuit for the given objective function. The 6 parameters in this circuit are θ[0],..., θ[5]. These are the classifier's trainable weights. Fig. 2 represents the quantum circuits using Ansatz. A precise initial assumption or informed estimate regarding the structure of a problem-solving solution is referred to as an ansatz in quantum machine learning (QML), and it is usually employed concerning variational algorithms. Ansatz refers to a parameterized quantum state or function that is thought to be a good approximation of the intended solution.

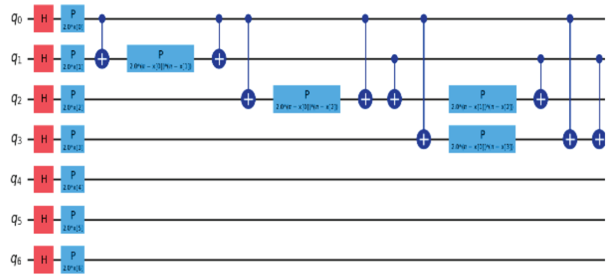

Fig. 1. Quantum Circuit using ZZFeatureMap.

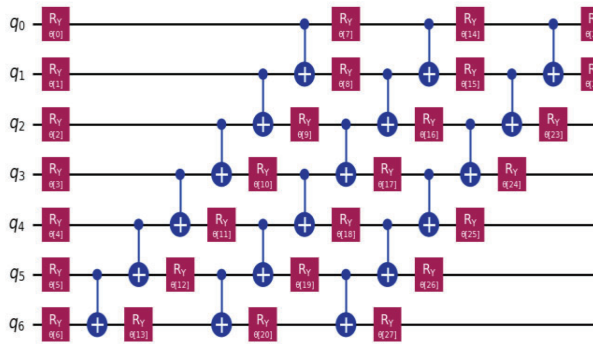

Fig. 2. Quantum Circuit using Anasz.

IV. Results and Discussion

The analysis and examination of two models, namely QSVM and classical SVM, were conducted to compare their speed and accuracy values. The dataset was split into 70% training and 30% testing ratios, just like any traditional machine learning algorithm. During data segmentation, the random sample selection method was used to make sure that the findings produced on the model's training and test data were reliable. Various feature maps and qubit counts were employed in the QSVM implementation to provide a more thorough analysis. Python programming was utilized to create the study's algorithms. Several libraries under the Qiskit library as well as the Qiskit library itself were utilized to apply quantum machine learning techniques. Depending on the quantity of each feature, both classical and quantum machine learning approaches were used in the IBM Quantum Lab environment. It can be seen that as the number of characteristics in the SVM algorithm increases, accuracy falls. Table 2 displays the classification performance metrics findings for the five characteristics.

The association between the problem variables is displayed in the correlation map in Fig. 3. To measure the monotonic link between variables that represent various mental illnesses, this correlation uses Pearsons' correlation coefficient. The correlations that have been identified between mental health illnesses have shed light on common risk factors or underlying vulnerabilities that may be responsible for their co-occurrence. Individuals may be predisposed to various psychiatric illnesses by common genetic, environmental, and psychological variables, resulting in patterns of comorbidity seen in the population. This enables them to customize treatment programs that address multiple mental symptoms and enhance overall results. In this work, the quantum-based machine learning algorithm QSVM was employed. Feature maps were used to transform the bits into qubits to use this approach. Five distinct feature maps were employed twice during this process. The distribution plot of the target variable is shown in Fig. 4.

Fig. 3. Correlation between reduced features.

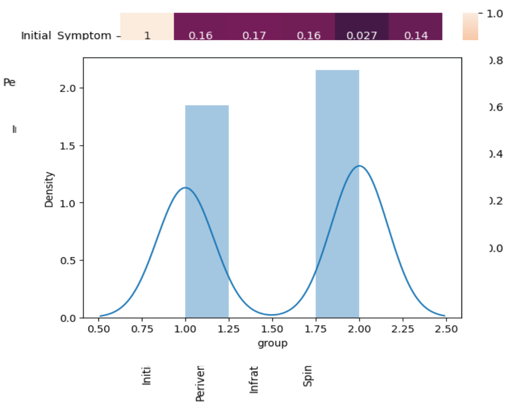

Fig. 4. Distribution plot of the target variable.

In a machine learning classifier, analyzing the distribution plot of the target variable is crucial for understanding the balance of classes within the dataset. This plot visually represents how the instances are distributed across different classes, helping identify any class imbalance that could impact model performance. Both the classes have been shown to follow a normal distribution. This shows that the model can learn equally from all classes, leading to potentially better generalization.

Fig.5 displays the objective function value against each QSVM iteration using RealAmplitutde method of QSVM classifier. It is evident that about 15 iterations were required to determine the ideal value of the objective function. On the other hand, the EfficientSU4 algorithm required 40 seconds to generate these results in around 25 iterations as shown in Fig.6. The method becomes less difficult and converges more quickly after performing feature reduction and optimization. The results obtained from the classical SVM and QSVM algorithms are provided in Table II.

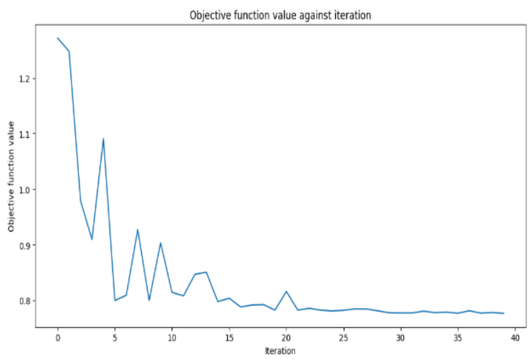

Fig. 5. Objective Function value of RealAplitude algorithm.

Fig. 6. Objective Function value of EfficientSU2 algorithm.

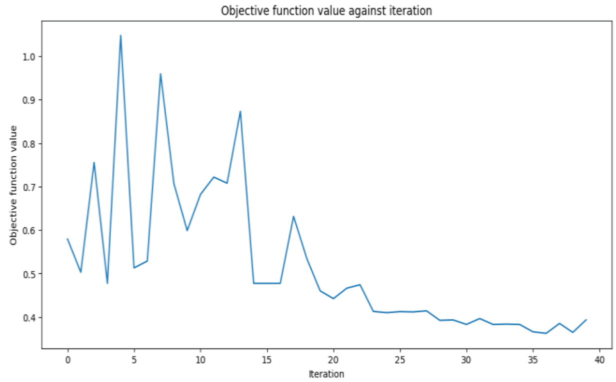

TABLE II. PERFORMANCE ANALYSIS

Prediction Model	Accuracy (%)	F1-score	Time (in secs.)
QSVM (before feature optimization)	72	0.75	294
QSVM (after feature optimization) with ZZ feature maps (11-15 qbits)	80	0.82	40
QSVM - Real Amplitudes (4 qbits)	94	0.94	35
QSVM - EfficientSU2 (4 qbits)	96	0.97	35

It is evident from looking at the table that ZZ features maps work better when the entanglement is linear as opposed to full. With the exception of the Z and X feature maps, QSVM's accuracy rate for 11–15 qubits is quite poor. For four qubits, the best performance is attained. All qubit values saw 100% performance from the X feature map. A speed reduction is seen across different feature maps starting at 5 qubits.

Another finding is that better outcomes can be obtained with even a small adjustment to the ansatz. Compared to the RealAmplitudes model, the two-feature model that uses the EfficientSU2 ansatz performs better.

V. CONCLUSION

This study compares and evaluates computational techniques with traditional techniques for predicting MS. This study used the MS dataset to achieve high performance in binary classification. The proposed approach carried out advanced optimized parameters and methodologies to achieve high performance. The proposed model achieves higher accuracy by adding computational tools. This will lead to reduced processing time and is probable for practical applications in medical data analysis. In this research, the effective inclusion of preprocessing methods will lead to high precision and computational efficiency. The results lead to addressing complex classification problems. Future studies focus on extending work to handle multi-class classification problems. This study will improve the effect of methodologies in healthcare applications.

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